

Homework 7

ORF 245 - Fundamentals of Statistics

Due: Apr. 11th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Simulation: CI's in R)

In this exercise, you will construct in R many distinct realizations of confidence intervals for an unknown mean, and return the fraction of intervals containing the true mean.

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where the value of μ is known, and σ^2 is assumed to be known.

(a) In R, write a piece of code that does the following:

INPUTS:

- Population mean μ (real number)
- Population variance σ^2 (positive real number)
- Sample size n (positive integer)
- Confidence level α (in $(0, 1)$)
- Number of trials M (positive integer)

PROCEDURE:

- For each trial in $1, 2, \dots, M$:
 - Generate realizations $x_1, \dots, x_n$ from a $N(\mu, \sigma^2)$ distribution.
 - Use $x_1, \dots, x_n$ to compute a realization of the two-sided $1 - \alpha$ confidence interval for the population mean, given by $\bar{x}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.
Note: for the critical values of a standard normal distribution, use the `qnorm()` function.
 - Check whether this interval contains μ .
- Return the fraction of the M trials in which the realized $1 - \alpha$ confidence interval contained the true population mean.

Note: For this part of the exercise, you need only submit your code – a screenshot will suffice.

Code used in part (a):

```
simulate_CI <- function(mu, sigma2, n, alpha, M) {  
  sigma <- sqrt(sigma2)  
  z <- qnorm(1 - alpha / 2)  
  count <- 0  
  
  for (i in 1:M) {  
    x <- rnorm(n, mean = mu, sd = sigma)  
    x_bar <- mean(x)  
    margin <- z * sigma / sqrt(n)  
    lower <- x_bar - margin  
    upper <- x_bar + margin  
    if (mu >= lower && mu <= upper) {  
      count <- count + 1  
    }  
  }  
  
  return(count / M)  
}
```

- **Solution:** We implemented the simulation procedure using the above R function. This function runs M trials, each time computing a $1 - \alpha$ confidence interval of the form

$$\bar{x}_n \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

and returns the fraction of intervals that contain the true mean μ .

- (b) Let $\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{4}$, $n = 10$, $\alpha = 5\%$, and $M = 1000$, and run your procedure. Report the fraction of confidence intervals that contained the true mean. What do you see? Why?

- **Solution:** We ran the R function with the following parameters:

$$\mu = \frac{1}{2}, \quad \sigma^2 = \frac{1}{4}, \quad n = 10, \quad \alpha = 0.05, \quad M = 1000$$

The result was a fraction of confidence intervals containing the true mean equal to approximately 0.948.

This is close to the nominal confidence level of 95%. The small deviation is due to sampling variability inherent in the simulation. Overall, the confidence interval performs as expected.

- (c) Repeat the same exercise, but with $n = 100$. What, if anything, changes? Why?

- **Solution:** We repeated the simulation with $n = 100$ and obtained a coverage rate of approximately 0.952.

Since the data are normally distributed with known variance, the theoretical confidence level of the interval

$$\bar{x}_n \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

is exactly $1 - \alpha = 0.95$. The slight deviation observed is due to randomness in the finite simulation sample ($M = 1000$). With n larger, the sample mean is more concentrated around the true mean, and the interval becomes narrower, but the coverage remains correct.

Exercise: 2. (CI's for Uniform Distribution Parameter)

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Unif}[0, \theta]$, each of whose density is given by:

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y := \frac{\max_{i=1, \dots, n} X_i}{\theta}$, whose density is given by:

$$f_Y(y) = \begin{cases} ny^{n-1} & 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Use $f_Y(y)$ to verify that

$$\mathbb{P}((\alpha/2)^{1/n} \leq Y \leq (1 - \alpha/2)^{1/n}) = 1 - \alpha$$

and use this statement to build a $1 - \alpha$ confidence interval for θ .

Hint: verify the statement by computing the CDF of Y .

- **Solution:** We start by computing the CDF of Y :

$$F_Y(y) = \int_0^y nt^{n-1} dt = y^n$$

So,

$$\mathbb{P}((\alpha/2)^{1/n} \leq Y \leq (1 - \alpha/2)^{1/n}) = (1 - \alpha/2) - \alpha/2 = 1 - \alpha$$

Now recall $Y = \max X_i / \theta$, which implies:

$$(\alpha/2)^{1/n} \leq \frac{\max X_i}{\theta} \leq (1 - \alpha/2)^{1/n}$$

Rewriting the inequality:

$$\frac{\max X_i}{(1 - \alpha/2)^{1/n}} \leq \theta \leq \frac{\max X_i}{(\alpha/2)^{1/n}}$$

So the two-sided $1 - \alpha$ confidence interval for θ is:

$$\left[\frac{\max X_i}{(1 - \alpha/2)^{1/n}}, \frac{\max X_i}{(\alpha/2)^{1/n}} \right]$$

(b) Verify that

$$\mathbb{P}(\alpha^{1/n} \leq Y \leq 1) = 1 - \alpha,$$

and build a $1 - \alpha$ confidence interval for θ based on this statement.

- **Solution:** We start by verifying the probability statement using the CDF $F_Y(y) = y^n$:

$$\mathbb{P}(\alpha^{1/n} \leq Y \leq 1) = F_Y(1) - F_Y(\alpha^{1/n}) = 1 - (\alpha^{1/n})^n = 1 - \alpha$$

Since $Y = \frac{\max X_i}{\theta}$, this implies:

$$\mathbb{P}\left(\alpha^{1/n} \leq \frac{\max X_i}{\theta} \leq 1\right) = 1 - \alpha \Rightarrow \mathbb{P}\left(\max X_i \leq \theta \leq \frac{\max X_i}{\alpha^{1/n}}\right) = 1 - \alpha$$

Thus, a $1 - \alpha$ confidence interval for θ is:

$$\left[\max X_i, \frac{\max X_i}{\alpha^{1/n}} \right]$$

Exercise: 3. (CI's for Exponential Distribution Parameter)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$, for some unknown parameter $\lambda > 0$. Assume that $n > 40$. We will build two distinct $1 - \alpha$ confidence intervals for the parameter λ .

- (a) Let $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

Show that for any given $\alpha \in (0, 1)$:

$$\mathbb{P} \left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \leq z_{\alpha/2} \right) = 1 - \alpha,$$

where $z_{\alpha/2}$ is the $\alpha/2$ critical value for a standard normal random variable.

- **Solution:** Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$. Then $\mathbb{E}[X_i] = 1/\lambda$ and $\text{Var}(X_i) = 1/\lambda^2$. By the Central Limit Theorem, for large n , we have:

$$\frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{\sigma} \approx \mathcal{N}(0, 1)$$

with $\sigma = 1/\lambda$.

Since λ is unknown, we approximate σ using the sample standard deviation s , giving:

$$\frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \approx \mathcal{N}(0, 1)$$

Therefore, we can write:

$$\mathbb{P} \left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \leq z_{\alpha/2} \right) \approx 1 - \alpha$$

This expression justifies the probabilistic statement used for constructing confidence intervals for λ . Note that, for any given $\alpha > 0 \in (0, 1)$, z_α satisfies $\mathbb{P}(Z > z_\alpha) = \alpha$ and we assume that $z_\alpha \geq 0$.

- (b) Use the expression in part (a) to construct a two-sided $1 - \alpha$ confidence interval for λ . Be careful to notice that λ is **not** the mean of $X_1, \dots, X_n$.

- **Solution:** From part (a), we have the approximate relation:

$$\mathbb{P} \left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \leq z_{\alpha/2} \right) \approx 1 - \alpha$$

Solving this inequality for $\frac{1}{\lambda}$, we get:

$$\bar{X}_n - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \leq \frac{1}{\lambda} \leq \bar{X}_n + z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Note that since the r.v.s X_i are exponentially distributed, the sample mean $\bar{X}_n$ is always positive. Therefore, $\bar{X}_n + z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$ is also always positive.

Next, inverting all sides yields a two-sided $1 - \alpha$ confidence interval for λ :

$$\left[\frac{1}{\bar{X}_n + z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}}, \frac{1}{\bar{X}_n - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}} \right]$$

- (c) Build a **lower** one-sided $1 - \alpha$ confidence interval for λ . Be sure to begin with a probabilistic statement (as in part (a)), justify it, and show your work in obtaining the confidence interval. Your interval for λ should be of the form $[\cdot, \infty)$.

• **Solution:** We begin with the one-sided probabilistic inequality:

$$\mathbb{P} \left(\frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \geq -z_\alpha \right) \approx 1 - \alpha$$

Solving this inequality:

$$\frac{\sqrt{n}(\bar{X}_n - \frac{1}{\lambda})}{s} \geq -z_\alpha \Rightarrow \bar{X}_n - \frac{1}{\lambda} \geq -z_\alpha \cdot \frac{s}{\sqrt{n}} \Rightarrow \frac{1}{\lambda} \leq \bar{X}_n + z_\alpha \cdot \frac{s}{\sqrt{n}}$$

Thus, the lower bound for λ is:

$$\lambda \geq \frac{1}{\bar{X}_n + z_\alpha \cdot \frac{s}{\sqrt{n}}}$$

The final one-sided $1 - \alpha$ confidence interval for λ is:

$$\left[\frac{1}{\bar{X}_n + z_\alpha \cdot \frac{s}{\sqrt{n}}}, \infty \right)$$

Exercise: 4. (Setting up Hypothesis Tests)

Let $X_1, \dots, X_n$ be iid random variables with mean μ and variance σ^2 . Moreover, let $\bar{X}_n$ be the (random) sample mean and $s^2(X_1, \dots, X_n)$ be the (random) sample standard deviation.

For each of the following assertions, state whether it is a legitimate set of hypotheses. If they are not legitimate hypotheses, justify why not.

- (a) $H_0 : \mu = 1$ and $H_A : \mu > 1$.

• **Solution:** This is a valid hypothesis test. Both the null and alternative concern the population mean μ , which is a fixed (but unknown) parameter. The null asserts $\mu = 1$, and the alternative asserts $\mu > 1$, making it a one-sided test.

- (b) $H_0 : \frac{\mu}{\sigma} = 1$ and $H_A : \frac{\mu}{\sigma} > 1$.

- **Solution:** This is a valid hypothesis test. The quantity $\frac{\mu}{\sigma}$ involves two population parameters. Hypothesis tests can target such functions of parameters, although the test procedure may be more complicated. The logic of the null vs. alternative is valid.

(c) $H_0 : \bar{X}_n = 1$ and $H_A : \bar{X}_n > 1$.

- **Solution:** This is not a valid hypothesis test. The null and alternative hypotheses refer to the sample mean $\bar{X}_n$, which is a random variable and not a fixed parameter. Hypothesis tests must concern population parameters (like μ), not sample statistics.

(d) $H_0 : \frac{\bar{X}_n - \mu}{s/\sqrt{n}} = 1$ and $H_A : \frac{\bar{X}_n - \mu}{s/\sqrt{n}} > 1$.

- **Solution:** This is not a valid hypothesis test. The null and alternative concern the value of a test statistic, which is a function of sample data and is random. A proper hypothesis test would test something like $H_0 : \mu = \mu_0$, and then use this statistic to perform the test, not to define the hypotheses themselves.

Exercise: 5. [OPTIONAL] (Two-Sample Confidence Intervals)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu_X, \sigma_X^2)$ and $Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu_Y, \sigma_Y^2)$, such that all of the random variables $X_1, \dots, X_n, Y_1, \dots, Y_n$ are mutually independent.

- (a) What is the distribution of the difference $X_1 - Y_1$? Be sure to state and justify: i) the family of distribution, and ii) any relevant parameters.

Hint: Recall that all of these random variables are mutually independent.

- **Solution:** Since $X_1 \sim N(\mu_X, \sigma_X^2)$ and $Y_1 \sim N(\mu_Y, \sigma_Y^2)$, and the two are independent, their difference is also normally distributed:

$$X_1 - Y_1 \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

This follows from the properties of the normal distribution under linear transformations and independence.

- (b) What is the distribution of the *average* difference $\frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$?

- **Solution:** Let $D_i := X_i - Y_i$. Then $D_i \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$, and the D_i are iid.

Thus, the average difference $\bar{D}_n := \frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$ satisfies:

$$\bar{D}_n \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2 + \sigma_Y^2}{n}\right)$$

- (c) Use the result in part (b) to build a two-sided $1 - \alpha$ confidence interval for the *difference* in means, $\mu_X - \mu_Y$. Be sure to: i) start with a justified probabilistic statement, and ii) show every step in deriving your confidence interval.

• **Solution:**

Let $D_i := X_i - Y_i$, and recall from part (b) that $D_i \stackrel{\text{iid}}{\sim} N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$. Then the sample mean $\bar{D}_n = \frac{1}{n} \sum_{i=1}^n D_i$ is an estimator for $\mu_X - \mu_Y$.

Let $s_D^2 := \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D}_n)^2$ denote the sample variance of the differences. By the Central Limit Theorem, we have:

$$\frac{\bar{D}_n - (\mu_X - \mu_Y)}{s_D / \sqrt{n}} \approx \mathcal{N}(0, 1)$$

Therefore, a two-sided $(1 - \alpha)$ confidence interval for $\mu_X - \mu_Y$ is:

$$\left[\bar{D}_n - z_{\alpha/2} \cdot \frac{s_D}{\sqrt{n}}, \quad \bar{D}_n + z_{\alpha/2} \cdot \frac{s_D}{\sqrt{n}} \right]$$